

5-Year Experiment Report — System One

System One-with-the-Market — 5-Year Experiment Report

Repo: `github.com/manshu07/System One-with-the-Market` · **Branch:** `5yr-experiment` · **Window:** 2021-10-01 → 2026-09-22 (1,223 trading sessions)

Prepared for: Himanshu. Covers the 5-year extension of the “Can System One Invest?” experiment: data, repairs, monthly returns, trade holding periods, and the System One decision-model status.

Verification basis: every number below comes from scripts executed on this machine against the downloaded Yahoo Finance cache and the repo’s own simulation engine. Evidence files: `results/phase5y_free.md`, `results/monthly-analytics.{md,json}`, `results/nse-repairs.{md,json}`, `results/phase1_data_quality.md`.

1. Executive summary

Strategy (₹10,00,000 each, same rules)	Final value	Total return	Max drawdown	Trades
Momentum (top-5 by 20-day return, rebalanced daily)	₹23,76,000	+137.60%	-47.9%	1,432
NIFTY 100 buy-and- hold	₹13,76,364	+37.64%	-17.6%	1 (synthetic)
Random (seeded, same churn)	₹78,167	-92.18%	-92.4%	~2,900

- Momentum beat the index in **30 of 60 months** and was positive in 28/60 — the edge is real but concentrated: 2022 **+44.2%** and 2023 **+33.5%** did the heavy lifting; 2025 was **-11.8%** and 2026 (9 months) -4.7%.
- **Trade holding periods:** average **4.2 sessions**, median **2**, min 1, max 67. 1,129 of 1,427 closed trades lasted ≤ 5 sessions. This is a fast-churn system — and the random leg proves what churn costs when there is no signal: **-92%**.
- The **System One** decision-model leg is wired, smoke-proven on 5-year data (12/12 live calls, 118 ms mean), and awaiting a spending decision: ~96,000 paid gateway calls for the full run.

2. What was built (all committed on this branch)

Step	What happened	Result
1. Download	<code>config/data.json</code> widened to 2020-05-01 → 2026-09-23 (5y + ~105-session warm-up buffer beyond the 252 needed); <code>npm run download</code>	100/100 stocks cached, 1,585 sessions, 0 failures
2. Features	<code>npm run features</code> — 29 point-in-time indicators per stock-day	158,500 rows; decision-ready counts below
3. Data repair	Found Yahoo serving flat placeholder bars for 96/100 stocks on 2025-03-18 (live re-fetch proved it persists; NSE's official bhavcopy shows a fully traded session). Built <code>scripts/repair-from-bhavcopy.ts</code>	118 symbol-days patched from official NSE bhavcopy (incl. VEDL 2026-04-30 –65% bad print); 10 unreparable (symbol absent from bhavcopy EQ series) honestly logged
4. Rebuild	<code>npm run features</code> after repairs	Decision-ready rows: 95,499 → 122,328 (+28%). The 2025-03-18 poison had made every 200/252-session window null for ~a year — dead zone fully healed (93–98 ready tickers every month)
5. 5y simulation	<code>scripts/run-5y-free.ts</code> — repo's own engine (next-open execution, 10 bps cost, 5 bps slippage, max 5 positions, 20% cap), System One leg off	Full 1,223-session run + monthly/holding analytics
6. System One smoke	<code>scripts/jev-smoke-5y.ts</code> — live TypeScript API calls on 5y features, repo's exact prompt	12/12 OK , mean 118 ms, sane calibrated decisions (BUY 0.54–0.91, NO_ACTION 0.62–0.72)

Benchmark-gap handling: 6 of 1,229 sessions have a missing/invalid NIFTY 100 index bar while stocks traded (2022-12-26, 2024-01-01, 2024-02-19, 2025-01-01, 2025-02-01, 2026-01-01). Following the repo’s no-guessing philosophy, those sessions are **excluded and disclosed**, not interpolated.

3. Monthly returns (momentum vs NIFTY 100)

Full 60-month table with the random column lives in [results/monthly-analytics.md](#) . Highlights:

Year	Momentum	NIFTY 100	Months
2021 (Oct–Dec)	+6.0%	–3.3%	3
2022	+44.2%	+1.5%	12
2023	+33.5%	+17.0%	12
2024	+6.9%	+3.3%	12
2025	–11.8%	+7.6%	12
2026 (Jan–Sep)	–4.7%	–10.0%	9

- Best month: **+28.2%** (2022-11) · Worst month: **–12.9%** (2024-10)
- Beat NIFTY: 30/60 months · Positive months: 28/60
- Pattern: strong in trending markets (2022–23), bleeds in choppy/bear regimes (2025) — textbook momentum behaviour, now proven on this universe with full costs.

4. Holding periods — every trade’s days-after-buy

From the complete momentum ledger (1,432 trades, 1,427 closed):

Holding length (trading sessions)	Trades
1–5	1,129
6–10	149
11–20	117
21–40	28
41–60	3
60+	1

Average 4.2 sessions · Median 2 · Max 67. The rule mechanics explain it: a name is sold the day its 20-day return drops out of the top-5, so most positions live days, not months. The full trade-by-trade table (buy date, sell date, prices, P&L %, holding sessions for all 1,432 trades) is in [results/monthly-analytics.md](#).

5. What this proves — and what it doesn't

Proves: the pipeline scales to 5 years unchanged (phase separation paid off — only config + one repair script were needed); momentum on the current NIFTY 100 with realistic costs earned +137.6% over 5 years vs +37.6% buy-and-hold; Yahoo Finance needs an official-source repair layer for NSE (documented artifact class + working fix).

Doesn't prove: forward returns (survivorship bias — today's NIFTY 100 applied backward — flatters momentum more than the index because past winners are exactly the stocks that entered the index); System One's edge (that run hasn't happened yet — below); investability of a -47.9% max drawdown strategy at 10bps/5bps flat costs.

6. The System One leg – decision needed

	Detail
Status	Code path fully wired (<code>scripts/phase4-run.ts</code> , resume-safe, checkpointed); smoke-proven live on 5-year data (12/12 calls OK, 118 ms)
Cost	~122k sessions × ~95 eligible stocks ≈ 96,000 calls to the Vercel AI Gateway (<code>typesafe-ai/jev</code>) – paid; no gateway key on this box yet
Time	~16 h serial, ~4–5 h at concurrency 4 (backoff/retry machinery already built)
To start	Provide <code>AI_GATEWAY_API_KEY</code> → I set the new experiment id, run Phase 4, and this report gains the System One column + System One-specific trade holding stats

Not financial advice. Experimental backtest on repaired Yahoo/NSE daily data; flat cost assumptions; survivorship bias disclosed above.